CVRP: Quantum Clustering and Routing

DuQuantum / Pasqal | Track B | Local review draft | 25 October 2026

1. The complete pipeline and original Track B benchmark

We first group customers into vehicles, then choose visiting orders for those groups and evaluate total depot-to-depot distance. Each customer is served once, vehicle capacity is respected, and unused vehicles are allowed. Quantum evolution is emulated on a CPU; classical code fits geometry, checks capacity and decodes/selects measurements.

The formal benchmark contains 12 certified 2D instances: S/M/L, seeds 11/29, and uniform/heterogeneous demands. Customers lie in $[0,100]^2$; depot=(50,50). Five methods give 60 result rows. The original data and failed runs are retained; later slide comparisons and the 3D demo are separate experiments.

Track A input. A seeded window of at most eight customers proposes the next vehicle group. The score balances squared group sizes and normalized within-group distances, with lambda=0.15. Selected overloads and necessary remaining-load failures are rejected; the final vehicle receives the remainder if it fits. Capacity is postselected, not an exact analog penalty.

$$E_{\mathrm{cl}}/\kappa = \lambda m^2 - \mu m + \sum_{i<j} w_{ij} b_i b_j \qquad m = \sum_i b_i, \qquad \mu = 2\lambda \frac{|R|}{L}$$

Here b_i selects a customer, m is group size, R the remaining set, L the remaining fleet, and baseline kappa=4. Thirty-two trajectories reuse cached 1000-shot conditional banks. Routing uses 2000 measurements per eligible cluster. For fixed best-model groups, compare edge/global, edge/DMM, position/DMM and random permutations; also test random clustering/routing.

Group size	Formal routing action	Treatment
n<=3	Exact classical trivial route	Excluded from routing gaps/hits
n=4	Quantum batch or random comparison	Keep all valid and invalid shots
n>4	Declared exact classical size fallback	Counts in distance; lowers Q coverage

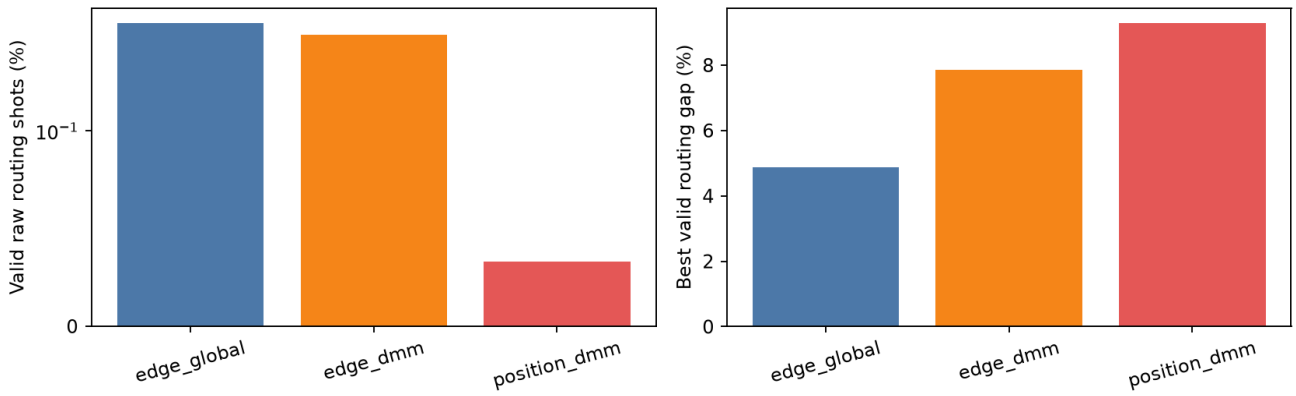


Figure 1. Original routing comparison. Left: mean raw valid-shot percentage, on a symlog scale. Right: mean best gap among batches with a valid route; failures are absent from that quality average, but remain failures in the table below.

Original routing variant	Feasible pipelines	Q calls	Successful Q groups
edge_global	11/12	15	15
edge_dmm	11/12	15	15
position_dmm	5/12	15	8

The original edge variants return full feasible plans on 11/12 instances; position/DMM returns 5/12. A failed four-customer quantum batch is not repaired classically. Full instance metrics, undefined gaps, coverage, pulses and raw counts remain in track_b_results/results.csv, clusters.csv and the saved banks.

2. Routing equations and the atom-energy mapping

Position model. z_{it}=1 places customer i at stop t, for t=1,...,n. L is the route-length term and P the penalty. The first penalty gives each customer one stop; the second puts one customer at each stop. In code, z/P/L are called x/A/C_{tour}; page 1 uses L for the remaining fleet, a different context.

$$E = L + P\sum_i\left(\sum_t z_{it} - 1\right)^2 + P\sum_t\left(\sum_i z_{it} - 1\right)^2$$

$$L = \underbrace{\sum_i c_{Oi} z_{i1}}_{\text{Depot to the first customer}} + \underbrace{\sum_{t=1}^{n-1}\sum_{i\neq j} c_{ij} z_{it} z_{j,t+1}}_{\text{Between adjacent positions}} + \underbrace{\sum_i c_{iO} z_{in}}_{\text{Last customer back to the depot}}$$

The three parts of L are depot-to-first, travel between consecutive stops, and last-to-depot. L has the units of c_{ij}; code normalizes travel costs by 100 sqrt(2), then evaluates reported routes in original distance units. Code indices 0,...,n-1 correspond to slide stops 1,...,n.

Edge model. e_{uv}=1 selects an undirected connection between locations u and v, including the depot. The objective adds selected edge lengths and penalizes any vertex degree other than two:

$$E = \sum_{u<v} \tilde{c_{uv}} e_{uv} + P\sum_v\left(\sum_{e\ni v} e - 2\right)^2$$

At four customers, position uses 16 atoms and edge uses 10. Code penalty A=P=10 on normalized costs exceeds the at-most-5 cost of a valid tour; a violated squared integer constraint costs at least 10. The full target is scaled by 0.1. Degree two implies one tour for 2-4 customers, but permits disjoint cycles at larger sizes; strict decoding checks connectivity. Exhaustive tests accept 12/1024 edge and 24/65536 position strings, verifying target algebra rather than physical sampling.

$$\frac{H(t)}{\hbar} = \frac{\Omega(t)}{2}\sum_i X_i - \sum_i \delta_i(t) n_i + \sum_{i<j}\frac{\Gamma}{r_{ij}^6} n_i n_j$$

$$h_i^{\text{real}} = -\delta_i(T), \qquad J_{ij}^{\text{real}} = \frac{\Gamma}{r_{ij}^6}$$

n_i counts Rydberg excitation, X_i mixes states, Omega drives transitions, delta is detuning, and Gamma=C_6/hbar. At final zero drive, positive detuning rewards excitation and Gamma/r^6 penalizes excited pairs. SLSQP fits all couplings, including unwanted interactions at desired zeros, with separation >=5 micrometers, radius <=38 and at most 16 atoms. Report relative pair-norm error. Global-only projects h to its mean; emulator-only DMM restores target linear costs but not pair couplings.

Routing uses an 800/3600/800 ns pulse (5200 ns), Omega_{max}=6 rad/us, initial delta=-12, final Omega=0. DMM ramps concurrently. QutipBackendV2 uses CPU state-vector evolution without noise, modulation response or calibrated layout. The exact assignment/slack construction belongs to Track A; it is not used as a full capacity Hamiltonian here.

3. CPLEX comparison and energy diagnostics

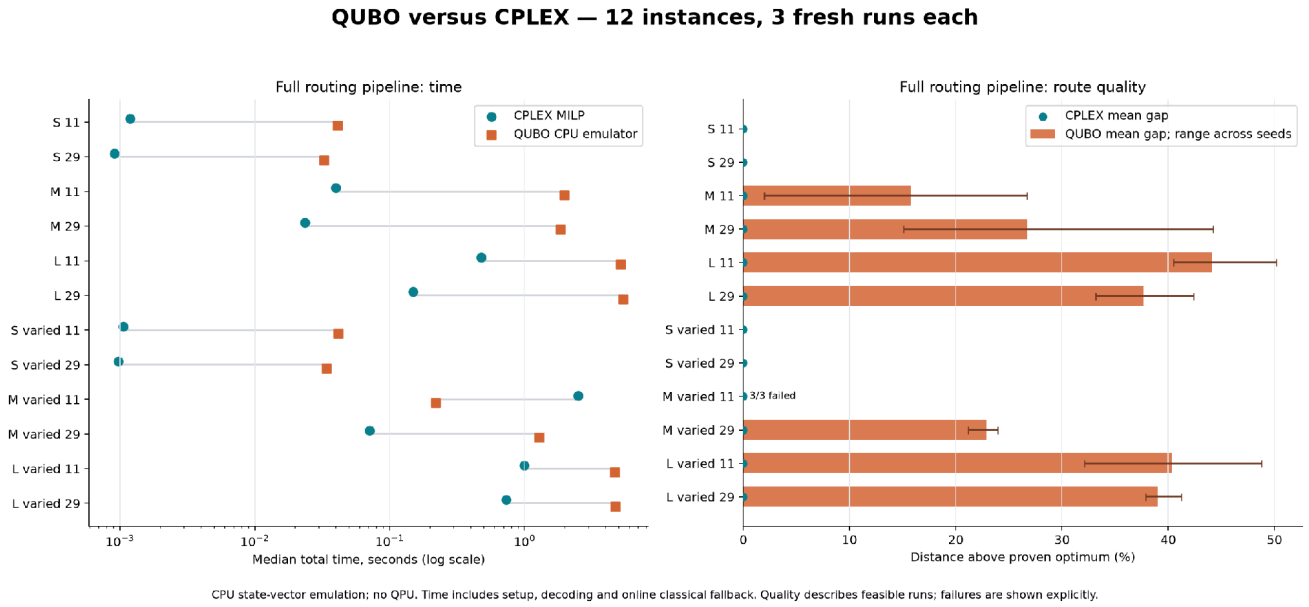


Figure 2. Independent classical comparison: 12 instances, three repeats each. Left: median total CPU time (log scale). Right: distance above the proven optimum; failed emulator runs remain marked. This comparison is distinct from Figure 1's original Track B experiment.

Full-route metric	CPLEX MILP	QUBO-assisted
Feasible runs	36/36	33/36
Mean route gap*	0%	20.59%
Median total CPU time	0.042 s	1.401 s
Proven optimal runs	36/36	0/36

*Mean gap covers 11 instances with successful emulator runs. All three failures are M_heterogeneous_11. CPLEX is faster overall and reaches the exact optimum; the CPU-emulated heuristic produces longer routes. A failed run can stop early, so its lower runtime is not a successful solve. These are not real-QPU timings.

Do the fitted atoms preserve the intended optimum?

Stage / encoding	Banks	Sampled target optimum	Native optimum overlap
Clustering / subset	754	707/754	0/754
Routing / edge	36	19/36	0/36
Routing / position	7	0/7	0/7

CPLEX and exhaustive enumeration agree on all 797 target-QUBO minima, but none of the 797 fitted native models retains a target-optimal bitstring as a ground state. Finite-time evolution can still sample useful excited states. Edge banks return valid tours in 36/36 cases, position in 3/7. These are diagnostic banks, not the 15-call original routing table.

$$\text{gap}(\%) = 100\left(\frac{C}{C^*} - 1\right)$$

For full-route gaps, C* is the exact CVRP optimum. For model gaps, use the exact F optimum. Failed outputs have undefined gaps. Routing gaps compare measured tours with exact tours on the same eligible groups; trivial and declared fallback groups are excluded. Raw validity and reference-hit rates count invalid shots in the denominator, with tolerance 0.000001 percent. Conditional route-hit products are not unconditional CVRP or best-of-batch success probabilities.

References enumerate capacity-feasible subsets, compute Held-Karp tours and solve exact anchored partition DP; vehicle-label symmetry is removed. S is checked by exhaustive assignments and constrained CPLEX (30 s, one thread, zero relative MIP gap). Exact tables supply declared classical routes and scoring. evaluate uses exact routed cost only to break equal-F ties, not to repair groups.

4. Atlas Orbit: the end-to-end website demonstration

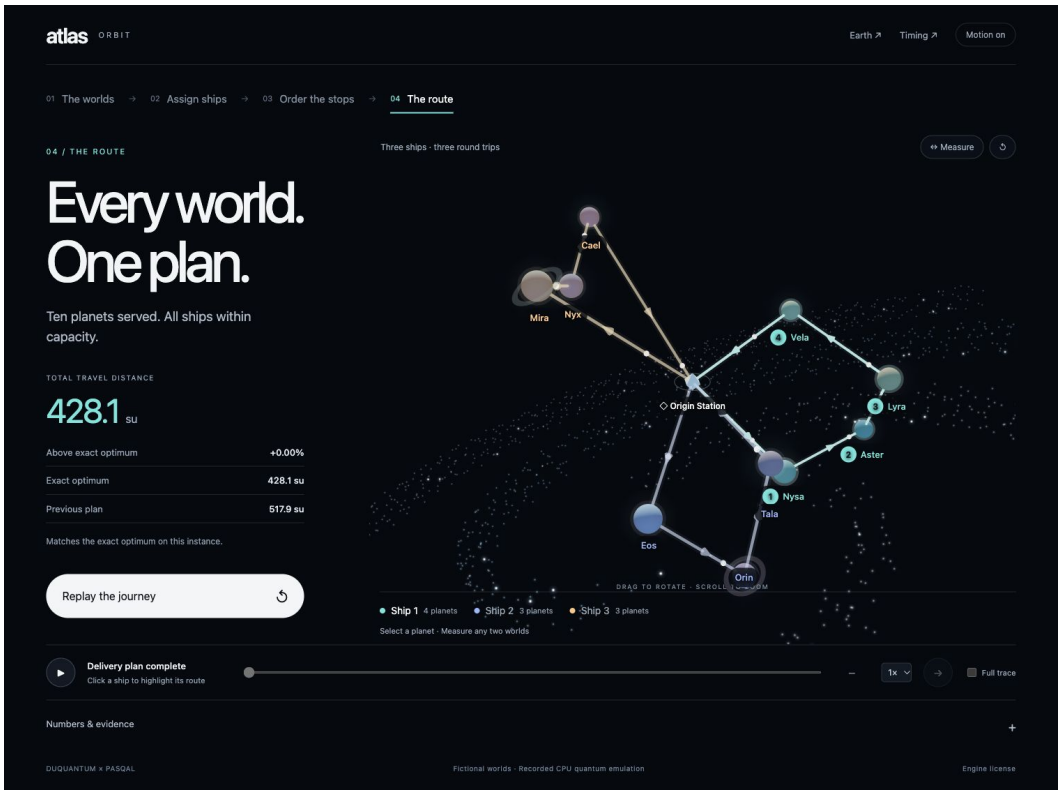


Figure 3. Atlas Orbit: ten fixed fictional planets, three ships and one station make the grouping and tour decisions visible. The website replays saved CPU-emulator evidence; it does not launch a quantum calculation in the browser.

Fixed-instance demo	Previous	Updated
Total distance	517.93 su	428.13 su
Gap above exact CVRP optimum	20.98%	0%
Clustering attempts / window orders	64 / 1	512 / 4
Routing shots per ship	2000	16000

The updated plan cuts travel by 17.34% and matches the exact optimum on this instance. Of 512 clustering attempts, 334 are feasible. All three tours come from edge/DMM samples with 10/6/6 atoms; capacities, decoding and selection remain classical. Larger budgets do not establish speedup or generalization. The 3D distance normalizer is 100 sqrt(3); atom registers remain 2D. Space units are fictional, not road distances or orbital flight paths.

The website follows inputs → ship assignment → visiting order → final routes. It displays capacity failures, measured candidates and the final total. Local replay (requires its server) | Website source and launch instructions.

$$C_{\text{end}} - C_{\text{ref}} = \underbrace{C_{\text{model}} - C_{\text{ref}}}_{\text{decomposition}} + \underbrace{C_{\text{shot}} - C_{\text{model}}}_{\text{clustering}} + \underbrace{C_{\text{end}} - C_{\text{shot}}}_{\text{routing}}$$

C_{ref} is the CVRP optimum; C_{model} exact-routes the F-optimal grouping; C_{shot} exact-routes the selected grouping; C_{end} uses the actual tours. The identity separates grouping-model, grouping-search and routing loss. Absolute distances add; the clustering term can be negative because F and travel differ.

What did not work. Correct target QUBOs did not ensure faithful geometry. Position batches often failed; DMM repaired linear costs but not pair distortion. The surrogate, sequential packing bounds, finite windows and shared-bank uncertainty remain limitations. The small-cluster random-permutation baseline is strong. Improve the model/embedding, then retest more independent seeds.

AI assisted formulation, coding, debugging and writing.

Sources: Pasqal challenge v3.0 (24 Oct 2026); team presentation (25 Oct 2026); track_b_results; SPACE_OPTIMIZATION.md; team CPLEX comparison and diagnostics; Lucas (2014), doi:10.3389/fphy.2014.00005; Pulser 1.9.1. Reproduction commands and pinned dependencies: README.md.